SENTRIX

Intelligent Multimodal
Home Security System

CAPSTONE PROJECT PROPOSAL · CPG NO. 299

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SENTRIX OVERVIEW

Sentrix is an **AI-powered smart security system** that uses **Video and Audio Intelligence** to detect threats in real time and trigger **Automated Emergency Responses.** It combines computer vision, behavioral analysis, and audio anomaly detection to accurately identify suspicious activities, verify identities, and understand context beyond basic motion detection.

The system integrates multiple AI detection engines into a unified **Threat Confidence Index (TCI),** which classifies threat levels and enables faster, smarter decision-making. By reducing false alarms and providing instant, actionable alerts, Sentrix ensures reliable, intelligent, and automated security for homes and premises.

COMPUTER VISION
YOLOv8 object & person detection

BEHAVIOURAL AI
Walk / Run / Crawl intent inference

AUDIO ANOMALY
Gunshot, scream, glass-break AI

IDENTITY ReID
Cross-camera person persistence

CLOUD THREAT
Weapon & fire override engine

AUTO DISPATCH
One-tap emergency authority alert

NEED ANALYSIS

1 RELIABLE THREAT VALIDATION

Reduce false alarms through multi-factor analysis

2 BEHAVIOR-BASED DETECTION

Identify suspicious intent, not just movement

3 CROSS-CAMERA IDENTITY TRACKING

Maintain continuous recognition across all feeds

4 MULTIMODAL THREAT DETECTION

Combine video + audio for higher accuracy

5 INSTANT EMERGENCY ACTIVATION

Enable one-tap or automated authority dispatch

6 ACCESSIBLE AI SECURITY

Deliver enterprise-level intelligence at consumer cost.

📖 LITERATURE SURVEY

A survey of existing work across the relevant technical domains

Real-Time Object Detection
YOLO architectures enable fast and accurate multi-object detection for live surveillance

Person Re-Identification (ReID)
Deep metric learning enables identity tracking across multiple non-overlapping cameras

Audio Event Classification
CNN models using mel-spectrograms accurately detect environmental threat sounds

IoT-Integrated Surveillance
Edge devices and cloud pipelines enable real-time monitoring and mobile alerts

Secure Evidence Storage
AES-GCM encryption and SHA-256 hashing ensure tamper-proof forensic evidence

Smart Surveillance in India
Low-cost edge AI solutions validated on Indian urban surveillance datasets

PROBLEM STATEMENT

Current home security systems rely on outdated designs and suffer from major limitations despite AI claims. There is **No Automated Bridge between Detection and Emergency response**, leaving users to act manually in critical situations. They produce extremely **High False Alarms** and lack behavioural intelligence to distinguish real threats from normal activity. They also fail to track identities across cameras and lack advanced audio threat detection.

Manual Emergency Alerts

No direct integration between detection and response

No Context Awareness

Systems detect motion, not intent or suspicious behavior

Identity Discontinuity

No persistent tracking of individuals across cameras

Audio Blind Spots

Critical sound-based threats are not recognized

False Alarm Overload

Majority of alerts are triggered by non-threatening events

Fragmented Intelligence

Vision, audio, and identity operate independently (if at all)

OBJECTIVES

Multimodal Threat Detection:
Real-time system combining vision, audio, and behaviour analysis to detect threats accurately

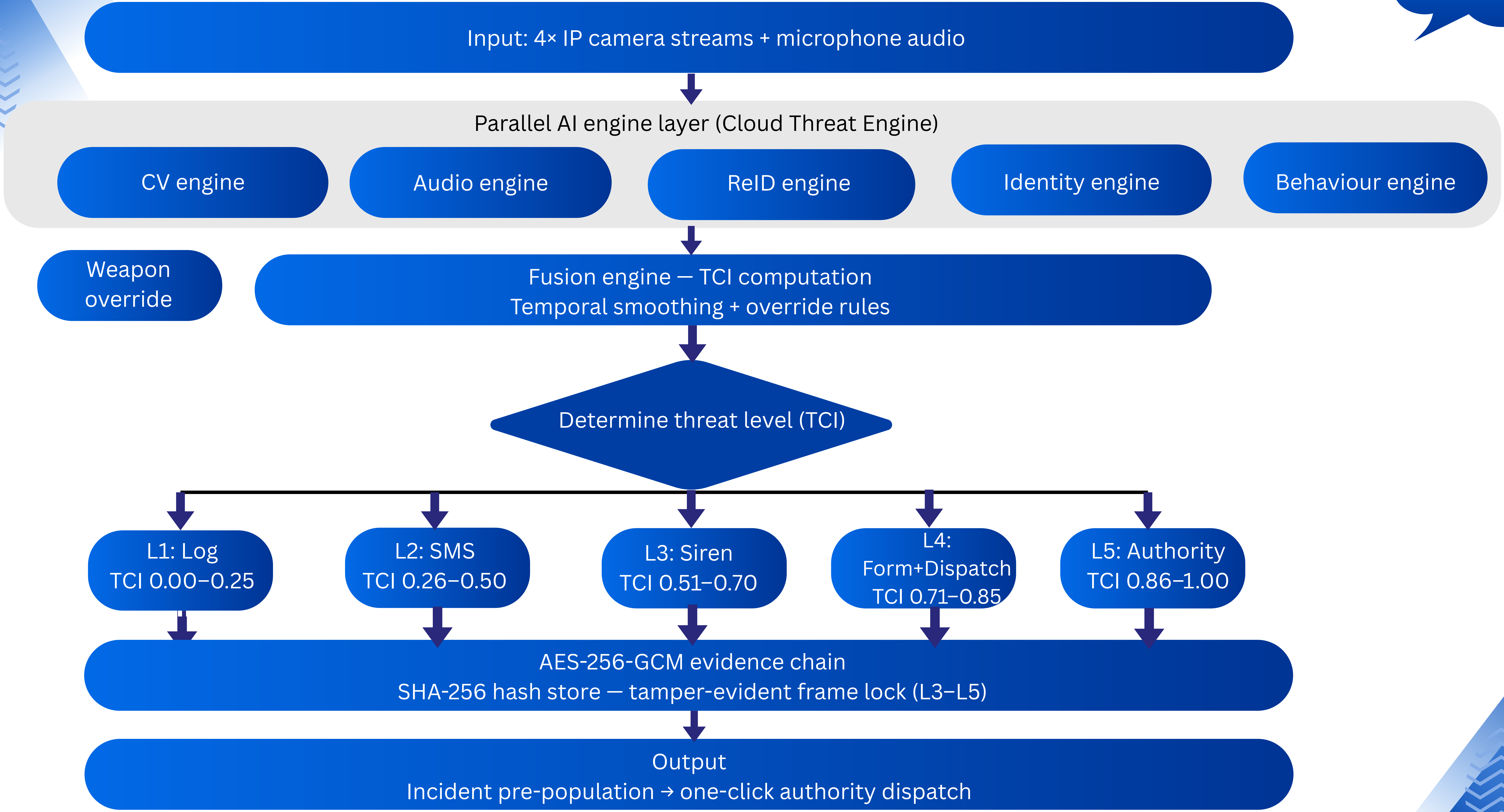
Cross-Camera Identity Tracking:
Maintains consistent identification of individuals across multiple camera feeds

Threat Scoring Framework (TCI):
Fuses inputs from all modules to generate a reliable threat level and response plan

Automated Emergency Response:
Mobile-integrated system for instant alerts, secure evidence handling, and quick authority dispatch

METHODOLOGY

ARCHITECTURE DIAGRAM



PROJECT REQUIREMENTS

HARDWARE

Component	Specification	Role in Sentrix
4× IP Cameras	1080p Full HD, PoE-compatible, night vision	Primary video input for CV, ReID, weapon & behaviour engines
Microphones (×2)	Directional / omnidirectional, 16 kHz sampling	Audio input for gunshot, scream, glass-break classification
Edge Processing Unit	NVIDIA Jetson Nano / Mini PC, ≥ 8 GB RAM, dedicated GPU	On-premise parallel execution of all 8 AI engines (<2 s latency)
PoE Network Switch	Gigabit PoE+, ≥ 4 ports	Powers and connects all cameras over single Ethernet run
Local Storage	SSD / NVR, ≥ 1 TB	AES-256-GCM encrypted evidence archive (Level 3–5 frames)
Mobile Device	Android / iOS smartphone	React Native dispatch app — push notifications & one-tap alert
UPS	≥ 30-minute backup	Maintains monitoring continuity during power outages

SOFTWARE

Library / Tool	Purpose in Sentrix
Python 3.10+	Core engine language — all AI modules and pipeline orchestration
YOLOv8 (Ultralytics) + OpenCV	Computer Vision Engine — person detection, pose estimation, weapon identification
TensorFlow / PyTorch	CNN model training and real-time inference for CV and audio engines
Librosa + AudioSet pipeline	MFCC spectrogram extraction and audio event classification (gunshot, scream, glass-break)
DeepSORT / OSNet	Person ReID and cross-camera identity persistence module
Face Recognition library	Identity Engine — enrolled visitor face matching against database
Twilio API	Automated SMS, voice call, and one-tap emergency authority dispatch
AES-256-GCM + SHA-256	Evidence encryption and tamper-evident frame hashing (Level 3–5)
React Native	Cross-platform mobile app — real-time dashboard and dispatch workflow
MQTT / WebSocket	Real-time IoT communication pipeline between edge unit and mobile app
PostgreSQL / SQLite	Storage for visitor profiles, TCI event logs, and evidence metadata

PROJECT OUTCOMES

Multi-Camera AI Platform

Real-time processing of 4 IP cameras at ≥15 FPS with <2s latency

FusionEngine Threat Scoring

Reduce false positives by ≥60% using multi-factor analysis

Audio Intelligence Engine

Detect gunshots, screams, and alarms with >90% accuracy

Mobile App & Emergency Dispatch

Real-time alerts with ≤10s authority dispatch

Cross-Camera Identity Tracking

Maintain consistent identity with ≥85% ReID accuracy

Secure Evidence Chain (AES-GCM)

Tamper-proof encrypted evidence with zero data compromise

WORK PLAN (GANTT CHART)

Sr. No.	Activity	Month	July		August				September				November				December			
		Week no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	Identification formulation and Planning of project	Plan																		
		Actual																		
2	CV Engine Development (YOLOv8, Pose, Weapon)	Plan																		
		Actual																		
3	Audio Intelligence Engine	Plan																		
		Actual																		
4	ReID & Identity Module	Plan																		
		Actual																		
5	FusionEngine & TCI Integration	Plan																		
		Actual																		
6	Mobile App & Emergency Dispatch	Plan																		
		Actual																		
7	System Integration & Field Testing	Plan																		
		Actual																		
8	Evaluation & Final Report	Plan																		
		Actual																		

INDIVIUAL ROLES

Kartik Garg — App Development

Akshay Ranveer — User Interface and Documentation

Prashant Gagneja — Core Machine Learning Implementation

Harshit Mishra — Core Machine Learning Implementation

Mehul Perimal — Hardware Development and Integration

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THANK YOU !